Nathan Ng

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Technical Skills

Programming Languages: Java, Python, Go, Kotlin, C/C++, JavaScript, SQL, HTML/CSS, Verilog (HDL)
Machine Learning: PyTorch, CNNs, NumPy, pandas, Scikit-Learn, Matplotlib, Jupyter, TPU Architecture
Hardware: FPGA Development, Digital Logic, Fixed-Point Arithmetic, Pipelining, Parallel Processing, Hardware Acceleration, HTF Architecture
Backend & Systems: REST APIs, React.js, FastAPI, Flask, Django, Node.js, AWS, PostgreSQL, Software Architecture, Ray Tracing, Performance Optimization
Development: Git/GitHub, Gradle, Docker, CI/CD, JUnit, Unit Testing, Test Automation, Linux, Fabric, OOP, Data Structures & Algorithms

Education

University of Ottawa Sep. 2025 – Dec. 2030
BASc in Software Engineering
- Digital Systems, Computer Architecture, Data Structures and Algorithms, Calculus 1-2, Linear Algebra, and more

Work Experience

Karate Instructor Oct. 2021 – Mar. 2025
Northern Karate Schools, Richmond Hill, ON
- Taught 300+ students, provided individualized feedback, communicated progress to parents, and helped maintain a 95% student retention rate.

Camp Counsellor Jun. 2022 – Aug. 2024
Medics Camp Canada, Richmond Hill & Markham, ON
- Worked with an 8+ person team to deliver STEM and first-aid programming, coordinate daily activities, and develop original educational games.

Projects

CHOOM - ML for Chinese Handwriting Recognition | *Python, PyTorch, NumPy, Scikit-Learn* Nov. 2025 – Present
- Developed a **Python/PyTorch CNN** for handwritten Chinese-character recognition using a dataset of over **1 million character images**.
- Built preprocessing, training, evaluation, and inference pipelines while comparing CNN architectures.
- Results from the long-term development of this project have since concluded, leading to the development of the FPGA accelerator.

CHM TPU (FPGA Implementation) | *HDL, FPGA, Digital Logic, Fixed-Point Arithmetic* Jul. 2026 – Present
- Currently implementing CHOOM's neural-network inference pipeline in **HDL (Verilog) for FPGA deployment for improved runtimes and minimized latency**.
- Designing dedicated hardware module with an instruction set for **convolution, multiply-accumulate, activation, pooling, activation (ReLu, Sigmoid), memory access, and control logic**.
- Applying **fixed-point arithmetic, pipelining, and parallel processing** and verifying hardware outputs against the Python reference model.

2TheMoon HTF Trading Program | *C/C++, Recurrent Neural Net (RNN)* Sep. 2026 – Present
- Designed, trained, tested and analyzed a **High trading frequency architecture model** that utilizes weight data from a RNN trained on pseudo-live trading data implemented by FPGA.
- Implemented C/C++ data processing programming to train the RNN using a physical TPU implemented by FPGA. **Opitimized code, reducing latency in order to optimize program.**
- Testing and applying RNN weight values to

SimuOttawa Team Lead – Ray Tracing & Optimization | *OpenGL, 3D Rendering, Optimization* Sep. 2025 – Present
- Leading development of the project's **ray tracing engine using OpenGL specification/API**, designed 3D engine system and component interfaces with modular workflow in mind.
- Developing and integrating **ray-tracing and rendering systems**, including, object selection, scene traversal, geometric intersection logic, lighting calculations, and performance-sensitive simulation components in a 3D rendering environment
- Lead engineer and maintainer of the SimuO website, designed the **software architecture in JavaScript using the React framework**.